Password Manager and Analyzer

Object Oriented Programming Lab (CSC-210)

BS(IT)-2(A)

Group Members

Name	Enrollment
1. MUHAMMAD HASAN	02-135251-040
2. SHAH RAIZ ASGHAR	02-135251-022
3. DANIYAL AFZAAL	02-135251-034

Submitted to:

Miss Madiha Aslam

BAHRIA UNIVERSITY KARACHI CAMPUS

Department of Computer Science

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1. Acknowledgment

I would like to express my sincere gratitude to our course instructor for providing us with the opportunity to work on this project, which helped us apply theoretical concepts of Object-Oriented Programming into a practical system. I am also thankful to my project teammate **Daniyal and Shahraiz** for his collaboration. Lastly, I appreciate the support of my peers and family who encouraged us throughout the development process.

2. Abstract

In the modern digital era, users are required to manage multiple passwords for different platforms, which often leads to weak password practices and security risks. This project, **Password Manager and Strength Analyzer**, is designed to help users securely store their credentials and analyze password strength. The system provides two major services: password strength analysis and a secure password vault. User authentication data is stored using a database, while encrypted user vault data is stored in text files. The application is developed using Java, following Object-Oriented Programming principles, and enhanced with a JavaFX-based graphical user interface.

3. Introduction

With the increasing use of online services, users are required to remember multiple usernames and passwords. Many users tend to reuse weak passwords, making them vulnerable to Cyber-attacks. A password manager helps users securely store and manage their credentials, while a password strength analyzer educates users about password weaknesses. This project aims to provide both functionalities in a single system using Java and modern software design practices.

4. Problem Statement

Users face difficulty in remembering multiple passwords and often compromise security by using weak or repeated passwords. Existing solutions are either complex or paid. There is a need for a simple, secure, and educational password management system that allows users to store passwords securely and understand password strength.

5. Existing Systems

Existing Systems	Limitations
Browser-based password managers	Limited control and transparency
Online password managers	No reminder about changing password within weak
Simple strength checkers	No weak password list added

How our system is different:

- Offline storage of user vault data
- Password encryption before storage
- Separate strength analysis feature
- Educational feedback on password weaknesses

6. Objectives and Goals

- To securely store user credentials
- To analyze password strength and highlight weaknesses
- To apply Object-Oriented Programming concepts
- To separate business logic from user interface
- To develop a scalable and maintainable system

7. Project Scope

The scope of this project includes:

- User registration and login
- Password encryption and decryption
- Secure password vault management
- Password strength analysis
- GUI-based interaction using JavaFX

The project does not include cloud storage or biometric authentication.

8. Overview of Project

The project is divided into multiple packages:

- `model` – Data representation classes
- `security` – Encryption and decryption
- `logindao` – Database connectivity and authentication
- `vaulthandler` – File-based vault operations
- `services` – Password strength analysis
- `utils` – Input and display utilities
- `gui` – JavaFX user interface

This layered structure improves maintainability and scalability.

9. Tools and Technologies

- **Programming Language:** Java
- **GUI Framework:** JavaFX
- **Database:** MS Access (via JDBC)
- **IDE:** IntelliJ IDEA
- **File Handling:** Text files
- **Encryption:** Custom cryptographic logic

10. Project Features

Functional Requirements

- User sign up and login
- Password encryption before storage
- Add, view, update, and delete passwords
- Password strength analysis

Non-Functional Requirements

- Security (encrypted passwords)
- Usability (GUI-based interaction)
- Maintainability (modular design)
- Scalability (future enhancements possible)

11. Course Implemented Concepts (Object-Oriented Programming)

Encapsulation

```
private String username;
private String passwordHash;
```

Abstraction

- DAO classes hide database operations
- VaultHandler hides file logic

Inheritance

- JavaFX classes extend `Application`

Polymorphism

```
@Override
public boolean equals(Object obj) {
    ...
}
```

Exception Handling

```
try {
    ...
} catch (IOException e) {
    ...
}
```

12. Output

-Main menu:

```
"C:\Program Files\Java\jdk-24\bin\java.exe" ...


    ╔═══════════════════════════════════════╗
    ║                                       ╕
    │      PASSWORD MANAGER & ANALYZER      │
    │                                       │
    │      Secure • Simple • Reliable       │
    ╒                                       ╕
    ╚═══════════════════════════════════════╝


  ╔═══════════════════════════╗
  ║                           ╕
  │  PASSWORD MANAGER & ANALYZER │
  ╚═══════════════════════════╝
1. Strength Analyzer
2. Password Manager
3. Close App

Enter your choice: |
```

-Strength analyzer

```
╔═══════════════════════════════╗
║  PASSWORD MANAGER & ANALYZER  ╕
╚═══════════════════════════════╝

1. Strength Analyzer
2. Password Manager
3. Close App

Enter your choice: 1
Enter your password: gshfgsu
Your password has the following weaknesses:
Password must be at least 8 characters long
Password must contain at least one uppercase letter (A-Z)
Password must contain at least one digit (0-9)
Password must contain at least one special character (!@#$%^&*()_+-=[]{}|;:',.<>?/)

Press Enter to continue...
```

-Login page:

```
╔═══════════════════════════╗
║       PASSWORD MANAGER    ╕
╚═══════════════════════════╝

1. Login
2. Signup
3. Back to Main Menu

Enter your choice: 1

╔═══════════════════════╗
║          LOGIN        ╕
╚═══════════════════════╝

Enter your username: daniyal
Enter your password: 123
Login successful! Welcome, daniyal

Press Enter to continue...
```

Activate Windows
Go to Settings to activate Windows.

-User dashboard:

```
╔═══════════════════════════════╗
║  Welcome, daniyal!            ║
╚═══════════════════════════════╝

1. Strength Analyzer
2. Password Manager
3. Sign Out

Enter your choice: 2


╔═══════════════════════════════╗
║        PASSWORD VAULT         ║
╚═══════════════════════════════╝

1. Add New Password
2. View All Passwords
3. Delete Password
4. Update Password
5. Back to Main Menu

Enter your choice:
```

Activate Windows
Go to Settings to activate Windows.

-Signup

```
╔═══════════════════════════╗
║           CREATE ACCOUNT          ║
╚═══════════════════════════╝

Enter your username: shahraiz
Enter your password: Raiz@_609
Your password has the following weaknesses:
Password is acceptable but recommended length is 12+ characters for better security
```

13. Future Work

- Cloud-based encrypted storage
- Two-factor authentication
- Password auto-generation
- Browser extension support
- Mobile application version

14. Conclusion

The **Password Manager and Strength Analyzer** project successfully demonstrates the practical implementation of Object-Oriented Programming concepts in Java. The system provides secure password storage and educates users about password strength. The modular architecture ensures that the project is maintainable, scalable, and suitable for future enhancements.